Unit 7: Legal and Ethical Aspects of Computing System

Complete exercise solutions — MCQs, Short Questions, Long Questions

EXERCISE SOLUTIONS — Unit 7: Legal and Ethical Aspects of Computing System

Multiple Choice Questions

1. Option typically included in common Terms of Use clauses is:

- a) User obligations
- b) Privacy and data use
- c) Product advertising
- d) Termination of service

Answer: a) User obligations

2. The purpose of the 'Limitations of Liability' clause in Terms of Use is:

- a) Ensures full company responsibility
- b) Limits company's liability
- c) Provides user rights for damages
- d) Guarantees constant service availability

Answer: b) Limits company's liability

3. The type of harmful software that secretly monitors user activity is:

- a) Spam
- b) Cookies
- c) Spyware
- d) Pharming

Answer: c) Spyware

4. The threat involving user redirection to fake websites is:

- a) Phishing
- b) Spam
- c) Spyware
- d) Pharming

Answer: d) Pharming

5. The impact of the digital divide on education is:

- a) Improves learning for all
- b) No educational impact
- c) Creates inequality in digital resource access

studysapiens.com

• d) Replaces traditional teaching methods

Answer: c) Creates inequality in digital resource access

6. The reason for understanding bias during information evaluation is:

• a) Ignoring other viewpoints

• b) Recognizing the source's agenda

• c) Validating all sources equally

• d) Avoiding evaluation altogether

Answer: b) Recognizing the source's agenda

7. The positive societal impact of computing systems is:

• a) Spread of misinformation

• b) Improved information accessibility

• c) Heightened privacy concerns

• d) Widened technology gap

Answer: b) Improved information accessibility

8. The practice considered ethical when using information is:

• a) Copying content without permission

• b) Respecting copyright and avoiding plagiarism

• c) Ignoring source attribution

• d) Using unverified information

Answer: b) Respecting copyright and avoiding plagiarism

9. The behavior considered responsible digital conduct is:

• a) Using secure websites

• b) Spreading false info online

• c) Respecting online privacy

• d) Reporting suspicious activity

Answer: d) Reporting suspicious activity

Short Questions

1. Why is it important for users to understand Terms of Use?

Understanding Terms of Use is important because: (1) it tells users what they are allowed and not allowed to do on a platform; (2) it explains how their personal data is collected, used, and shared; (3) it clarifies the provider's limitations of liability; (4) it protects users from unknowingly violating rules that could lead to account suspension. For example, a Careem user should know that a valid phone number is required and how their location data is used.

studysapiens.com

2. Differentiate between phishing and pharming.

Phishing is a scam where someone impersonates a trustworthy organisation — such as sending a fake bank email — to trick users into giving personal information like passwords or card numbers. Pharming redirects users to a fake website without their knowledge, even when they type the correct URL, by compromising DNS records or the host file. Key difference: phishing requires the user to actively provide information, while pharming works silently in the background. Both aim to steal credentials, but pharming is harder to detect because the address looks correct.

3. Identify two impacts of the digital divide on social and civic participation.

(1) Reduced access to government services — citizens without internet cannot use online portals for taxes, licences, or complaints, limiting their participation in civic life. (2) Limited access to information and public discourse — people without digital access cannot follow news, join online discussions, or express opinions, reducing their voice in society. In Pakistan, rural communities with limited internet struggle to access e-governance and online learning opportunities.

4. What are the key steps involved in evaluating information sources?

Key steps: (1) Check the source — is it a trusted organisation, academic journal, or known authority? (2) Check the author — are they qualified and credible? (3) Check the date — is the information current? (4) Check for bias — does the source have an agenda or conflict of interest? (5) Cross-check — verify the information against other reliable sources. (6) Check evidence — does the source provide facts, data, and references? These steps help distinguish reliable information from misinformation.

5. How does responsible data sharing contribute to ethical use of information?

Responsible data sharing means sharing information with consent, accuracy, and respect for privacy. It contributes to ethical use because: (1) it protects individuals' privacy and confidentiality; (2) it prevents the spread of misinformation by sharing verified content; (3) it respects intellectual property by giving credit to original sources; (4) it builds trust in digital communities. Sharing data responsibly ensures that information is used for good purposes without harming others.

Long Questions

1. Explain the common clauses found in Terms of Use and describe how they protect both the service provider and the user.

Common Terms of Use clauses: (1) User Obligations — define what users must do (e.g., provide accurate information, follow community standards). Protects the provider by setting rules, and the user by making expectations clear. (2) Limitations of Liability — restrict the provider's responsibility for service disruptions or losses. Protects the provider from excessive claims while informing users of the limits. (3) Privacy and Data Use — explain how personal data is collected and used. Protects the user's privacy rights. (4) Intellectual Property — state that content belongs to the provider or creators. Protects both parties' rights. (5) Termination of Service — allow either party to end the agreement. Together, these clauses create a fair framework that protects both sides.

2. Explain the concept of bridging the digital divide and discuss the roles of government initiatives in addressing this issue. Provide relevant examples from Pakistan.

Bridging the digital divide means closing the gap between those who have access to technology and those who do not. The divide is caused by economic barriers (unaffordable devices), geographical barriers (poor rural infrastructure), educational barriers (lack of digital literacy), and social barriers (age, gender, disability). Government initiatives in Pakistan: (1) Digital Pakistan programme — expanding internet access and digital services. (2) Telecommunication projects — laying fibre-optic cables in rural areas. (3) E-rozgar and IT training programmes — teaching digital skills. (4) Subsidised devices and internet packages for students. These initiatives improve access to education, employment, and government services for disadvantaged communities.

3. Explain how critical thinking skills contribute to responsible information utilization. Provide examples of how these skills can be applied in real-life scenarios.

Critical thinking helps users evaluate information before accepting or sharing it. It contributes to responsible use by: (1) identifying bias and agendas in sources; (2) verifying facts against multiple sources; (3) recognising logical fallacies and emotional manipulation; (4) avoiding the spread of misinformation. Real-life examples: (1) Before forwarding a WhatsApp message about a health cure, a critical thinker checks whether a reputable medical source confirms it. (2) When reading a news story, a critical thinker checks the author's credibility and whether other outlets report the same event. (3) Before making a purchase based on an online review, a critical thinker checks if the reviews are genuine or paid.

4. Discuss the importance of collaborative problem solving in a digital environment and provide examples of how human-machine collaboration can enhance this.

Collaborative problem solving in digital environments brings together human creativity and machine processing power. Its importance: (1) complex problems require diverse perspectives; (2) machines handle large-scale data analysis while humans provide judgement and context; (3) collaboration speeds up solutions. Examples of human-machine collaboration: (1) Doctors using AI diagnostic tools — the AI analyses scans, the doctor makes the final decision. (2) Teachers using adaptive learning platforms — the system personalises exercises while the teacher guides students. (3) Engineers using simulation software — the machine tests designs while humans refine them. This partnership produces better outcomes than either alone.